

Evaluation of Regional Charging Infrastructure Using Agent-Based Modelling

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Abstract. The transition to sustainable mobility in India is constrained by a co-evolutionary dilemma between electric vehicle (EV) adoption and charging infrastructure deployment: prospective buyers hesitate due to range anxiety, while station investors hesitate without guaranteed demand. This study develops a spatial simulation to evaluate how regional distribution network operators should time infrastructure rollout — reactively, in response to demand, or proactively, to stimulate it. The model integrates real-world geospatial data from OpenStreetMap (via the Nominatim and Overpass APIs) with household-level running-cost comparisons and a financial cost structure covering base setup, station hardware, annual expansion, and substation cabling costs. Household agents adopt EVs when the cost of charging (a flat ₹/km EV rate applied to distance from the nearest charging station) undercuts the cost of running on petrol (actual, year-specific ₹/km petrol prices applied to distance from the nearest petrol pump), with the catchment radius expanding each cycle to represent the station's growing service area over up to six annual simulation cycles (FY2023-24 onward). The model was applied to three Delhi localities — Lajpat Nagar, Rajouri Garden, and Chandni Chowk — each converging toward near-total EV adoption (99.7–100%) by year six, at a cumulative capital expenditure ranging from approximately ₹11.4 million to ₹12.6 million. The results suggest that proactive charging infrastructure investment can substantially accelerate regional EV adoption, with the pace of conversion strongly dependent on local household density, distance to competing petrol infrastructure, and the trajectory of real petrol prices relative to a fixed EV charging rate.

Keywords: Household Decision Simulation, Electric Vehicle Adoption, Charging Infrastructure Planning, Geospatial Simulation, Consumer Decision Modelling.

1 Introduction

1.1 Background

The global transition to sustainable mobility is hindered by a co-evolutionary “chicken-and-egg” dilemma regarding infrastructure deployment. Prospective buyers avoid electric vehicles due to severe range anxiety stemming from inadequate charging networks. Simultaneously, station investors and operators hesitate to commit capital without a guaranteed fleet to ensure utilization. This deadlock requires careful analysis of how consumer behavior and infrastructure availability interact over time.

1.2 Problem Statement

Distribution network operators and charging point operators (CPOs) in India have limited capital to deploy toward EV charging infrastructure and must decide *where* to invest it to maximize demand conversion per rupee spent. Locality characteristics — household density, existing petrol infrastructure proximity, and local fuel price trends — vary widely across Indian cities, meaning identical infrastructure spend can yield very different adoption outcomes depending on where it is placed. This study addresses the question: **for a fixed charging infrastructure investment, which locality profiles (by household density and competing petrol-pump proximity) yield the fastest and most cost-efficient EV adoption, and how should operators prioritize rollout across such localities to maximize demand generated per unit of capital?**

1.3 Objectives

This study utilizes a geospatial and economic simulation model to:

- **Simulate Spatially Driven EV Adoption:** Track the year-over-year conversion of households from non-EV (GREY) to EV-adopted (GREEN) across different Delhi localities based on their relative proximity to a centrally placed charging station versus the nearest existing petrol pump.

- **Evaluate Infrastructure Cost-Efficiency:** Calculate the total cost of charging station deployment over time—incorporating dynamic cabling costs from the nearest electrical substation, fixed hardware/setup costs, and annual recurring expansion costs—to determine the capital expenditure required per converted household.
- **Analyze Economic Tipping Points:** Model how historical and projected annual petrol prices (Rs/km) compared against a flat EV running cost (Rs 1.2/km) influence the rate at which households can financially justify switching to electric vehicles.
- **Optimize Locality Prioritization:** Compare adoption outcomes across various locality profiles to identify which environments (based on household density and petrol pump proximity) yield the fastest and most cost-effective EV transition per unit of capital invested.

2 Study Area and Scope

Study Area The geographical scope of this study focuses exclusively on the National Capital Territory of Delhi, India. The simulation evaluates various distinct urban localities within this region (such as Lajpat Nagar, Karol Bagh, and Connaught Place) to analyze how varying real-world infrastructure distributions impact localized EV adoption rates.

Scope

This project operates at the intersection of systems simulation, transportation infrastructure, and geospatial computing. The technical scope of the model is defined by:

- **Geospatial Topology Extraction:** Utilizing OpenStreetMap (OSM) data to dynamically pull real-world coordinates for residential buildings, petrol pumps, electrical substations, and public open areas within a specified radius of a locality's center.
- **Discrete-Time Adoption Dynamics:** Simulating localized EV adoption on an annual cycle, where non-EV (GREY) households transition to EV (GREEN) based on a cost-distance comparison between existing petrol infrastructure and the centrally placed charging station.
- **Financial & Deployment Modeling:** Evaluating the physical and financial requirements of station deployment, specifically calculating dynamic cabling costs (based on the distance from the nearest substation to the station's open-area placement) alongside fixed capital and annual recurring expenditures.

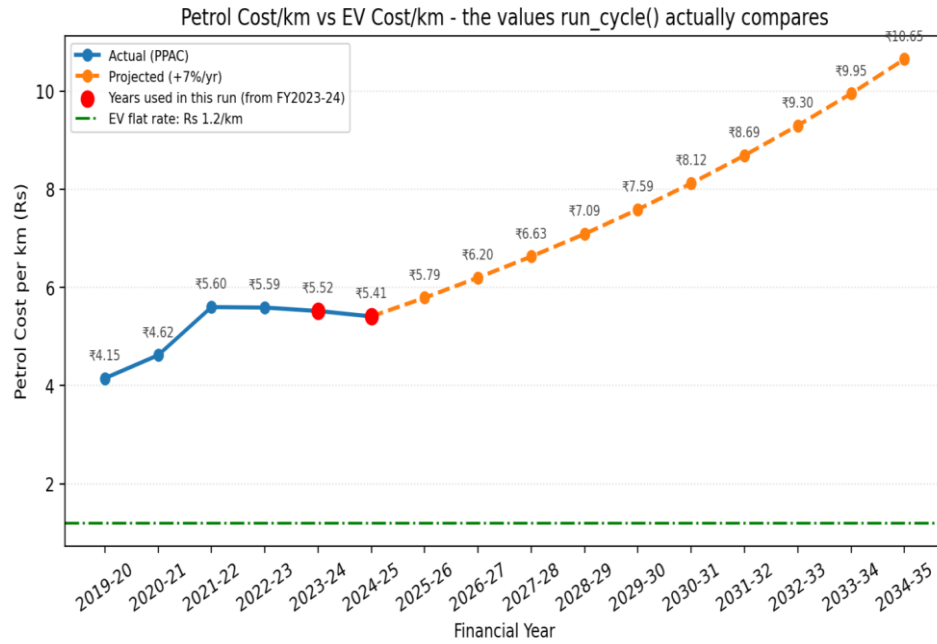
3 Data

3.1 Input Data for the Base Model

The model processes the following categories of input data to run a simulation cycle.

- **User-defined inputs:** Locality selection is made through a sidebar control that chooses a target neighborhood in Delhi from a predefined dropdown list (or a custom text input), which establishes a geographic center point via the Nominatim API. The number of simulation cycles (years) is set with a slider ranging from 1 to 6. This input directly dictates the maximum required geographic search radius for the final cycle and determines the cumulative duration for annual maintenance cost calculations.
- **Geospatial and system data (automated):** Based on the central coordinates, a dynamic fetch radius—capped at 5 km to optimize API performance—is used to query OpenStreetMap via the Overpass API. The query retrieves all residential buildings (with no artificial sample cap; every building in range is simulated), competing petrol pumps (`amenity="fuel"`), and infrastructure elements needed for placement. The charging station is strategically placed at the nearest open area (`leisure="park"`) to the nearest electrical substation (`power="substation"`), and the straight-line distance between these two points is measured to calculate the required cabling.
- **Behavioral baselines:** The simulation initializes with a synthetic baseline where a random 20% of the fetched households start as "GREEN" (already adopted EVs), while the remaining 80% begin as "GREY" (non-EV households).
- **Financial constants and dynamic costs:** Household conversion is driven by a comparative running-cost model. The running cost of EVs is fixed at Rs 1.2/km. This is weighed against dynamic, year-

specific petrol costs retrieved from a lookup table (e.g., starting at Rs 5.52/km for FY2023-24 and increasing annually based on actuals and a 7% projected growth rate). Fixed capital expenditures for deployment include a base setup fee of Rs 2,47,000, station hardware costs of Rs 33,00,000, an annual recurring expansion cost of Rs 13,00,000, and a cabling rate of Rs 2,800 per meter to wire the station to the local substation.



3.2 Output Data for the Base Model

The simulation generates two primary outputs: an interactive geospatial visualisation and longitudinal financial metrics. The geospatial output renders a dynamic map plotting the outward expansion of the geographic radius, charting the physical distribution of “GREEN” (EV-adopted) and “GREY” (non-adopted) households, competing petrol pumps, and the calculated cabling route to the nearest electrical substation. The financial output computes graphical metrics tracking cycle-over-cycle household conversions based on the distance multipliers. These adoption rates are cross-referenced against cumulative capital expenditure – including base infrastructure, hardware, and cabling costs – and validated against the continuous running-cost gate of ₹3/km for EVs versus ₹11/km for petrol.

Controls

Choose a Delhi locality

Rajouri Garden ▼

...or type a custom locality name (optional)

Number of cycles (years)

2

Run Simulation

Fig. 1. Interactive dashboard used to set the initial conditions of a simulation run (locality selection and number of simulation cycles).

Found **29093** houses and **38** petrol pumps in range. Charging station placed at the nearest open area to the nearest substation: **28.64421, 77.13737** (1.530 km from the geocoded center, 0.137 km of cable run from the substation).

Cycle 1 (FY 2023-24) Cycle 2 (FY 2024-25)

Fig. 2. Example inputs taken from Open map

4 Methodology

4.1 Global Parameters

Parameter Category	Variable / Constant	Assigned Value	Description / Constraint
Geospatial & Dimensional	Initial Search Radius	3 km	Starting a spatial radius around the charging station for Cycle 1 evaluation.
	Annual Radius Increment	+2 km	Expansion of the evaluation radius per simulation cycle (e.g., 3 km -> 5 km -> 7 km).
	Maximum Fetch Radius	5 km	Hard cap on Overpass API query radius to prevent server timeouts during data extraction.
	Max Residential Sample	None (Uncapped)	Every residential building returned by OpenStreetMap within range is fetched and

			simulated; rolled into groups of 50 for UI rendering.
	Cycle Unit	1 year	Discrete time step for each simulation iteration, starting at FY2023–24.
	Cycle Range	1 to 6 cycles	Allowable simulation duration slider range (representing 1 to 6 fiscal years).
Financial & Hardware	EV Running Cost	₹1.20 / km	Flat per-kilometer running cost used directly in the EV route distance calculation.
	Petrol Running Cost	₹5.41 – ₹10.65 / km	Fiscal year lookup table (PPAC actuals starting at ₹5.52 for FY23–24, projected at +7%/year thereafter).
	Base Setup Fee	₹2,47,000	One-time fixed administrative and site preparation fee.
	Station Hardware Cost	₹33,00,000	Capital expenditure for the core charging infrastructure hardware.
	Annual Expansion Cost	₹13,00,000	Ongoing yearly operational/expansion cost added to cumulative total per cycle.
	Cabling Cost Rate	₹2,800 / meter	Unit cost for running cabling from the nearest electrical substation to the station's park/open-area site.
Behavioral & Adoption	Initial EV Baseline	20%	Initial seed percentage of randomly assigned "GREEN" (already EV-adopted) households.
	Initial Non-EV Baseline	80%	Initial seed percentage of "GREY" (non-EV / potential adoptable) households.
	Cycle Decision Logic	Distance-Only (Cycle 1) Direct Cost Comparison (Cycle 2+)	Cycle 1: Pure spatial comparison (dstation < dpump). Cycle 2+: Direct cost comparison (Cost_EV < Cost_petrol with no artificial multipliers).

4.2 Decision Pathway for Simulation

Breakdown of the Logic

- **Initialization:** Before any cycles begin, the system shuffles all fetched houses and randomly assigns 20% of them to be **GREEN** (already EV-adopted), while the remaining 80% default to **GREY** (non-EV).
- **Geospatial Filtering:** In every cycle, the system only evaluates houses that fall strictly within the current cycle's expanding radius limit (starting at 3 km, growing by +2 km each year). Houses outside this radius are ignored for that year.
- **Cycle 1 (Distance-Only):** For the very first simulation cycle (representing FY2023-24), no financial cost multipliers are used. A GREY house simply converts to GREEN if the raw, straight-line distance to the charging station is shorter than the distance to the nearest petrol pump. (taken only because initial EV demand growth shall be seen only in immediate vicinity of a charging station due to limited charging points, which is lifted as cycles go by because of expansion costs being included)
- **Cycle 2+ (Cost Comparison):** From the second cycle onwards, the pure distance comparison is replaced by a financial cost model. The script multiplies the charging station distance by a flat ₹1.2/km and the petrol pump distance by the specific fiscal year's petrol rate. The house converts to GREEN only if the EV route evaluates to be cheaper than the petrol route.

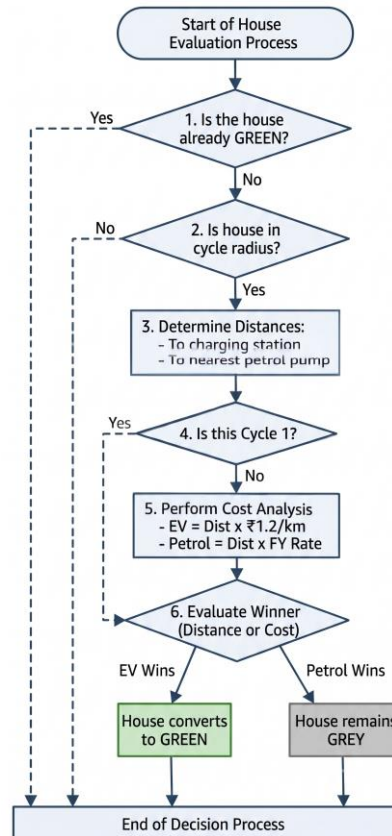


Fig. 2. Consumer decision-making pathway followed by each household agent in every simulation cycle

Important- The choice of multiplying distance by respective costs of both decision pathways has been taken only because the placement strategically can only optimize this element of EV ownership, day to day usage costs, catering to EV ownership catalyzes EV demand, is the base assumption that has been made.

Note - Once a grey dot is taking a decision, range-wise, the 50 houses it represents check if the majority has turned green or not, if they have, they convert, so each dot follows a majority and minority system whilst checking if they should stay grey or turn green.

5 Results and Discussion

5.1 Base Model Results

A two-cycle base model run illustrates the adoption and cost dynamics produced by the simulation.(DWARKA)

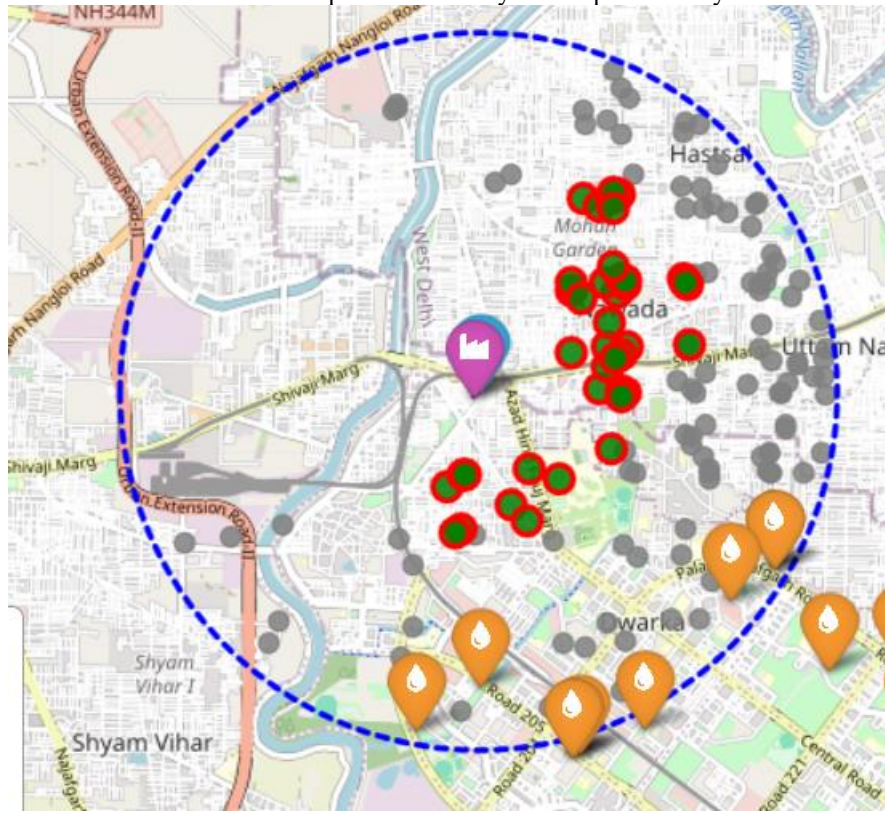


Fig 4. The placement of all blips, where yellow blips represent petrol pumps, grey dots are non-EV households, circled green houses are freshly converted houses,

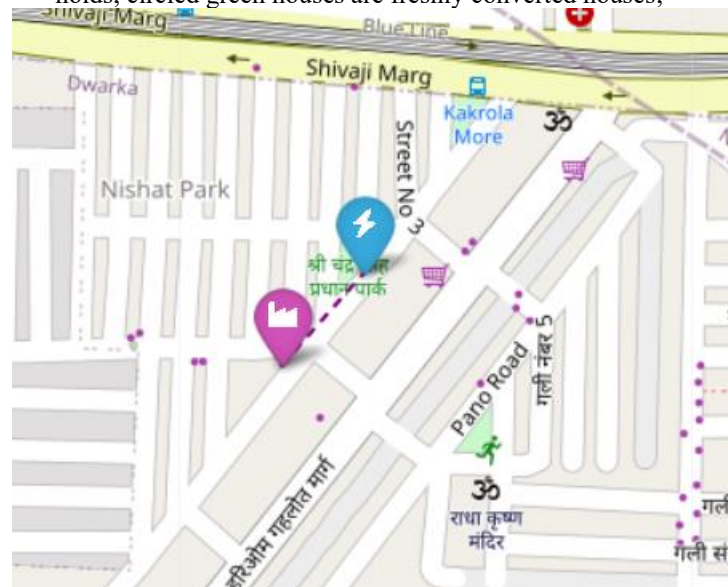
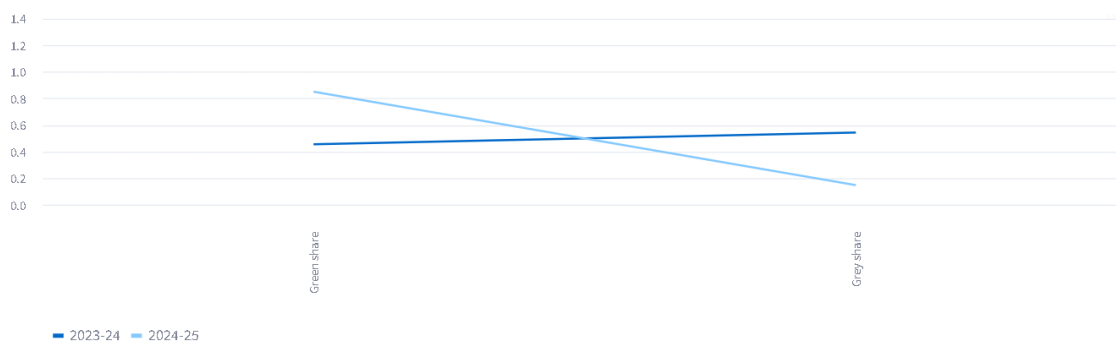


Fig 5. The placement of charging station strategically takes place as near a possible to a central substation.

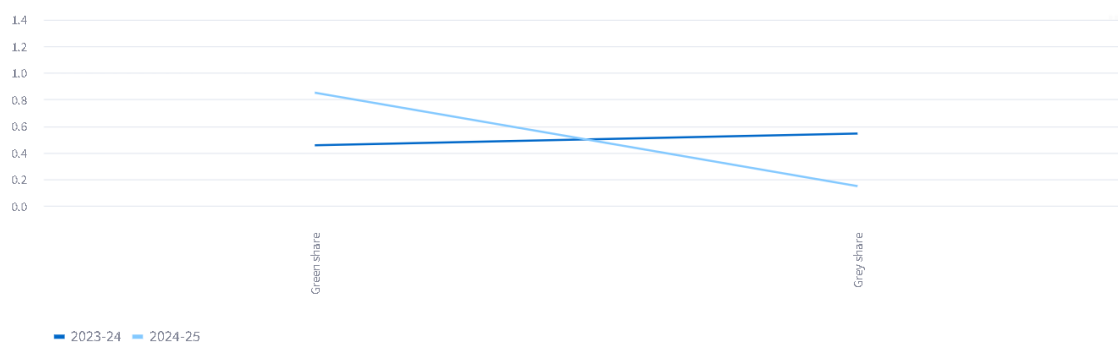
Share of houses (grey falling, green rising, as a fraction of all houses within radius)



Cost breakdown by year

Year	Expansion cost this year (Rs)	Cumulative cost / maintenance increment (Rs)
1	1300000	5028890.044
2	1300000	6328890.044

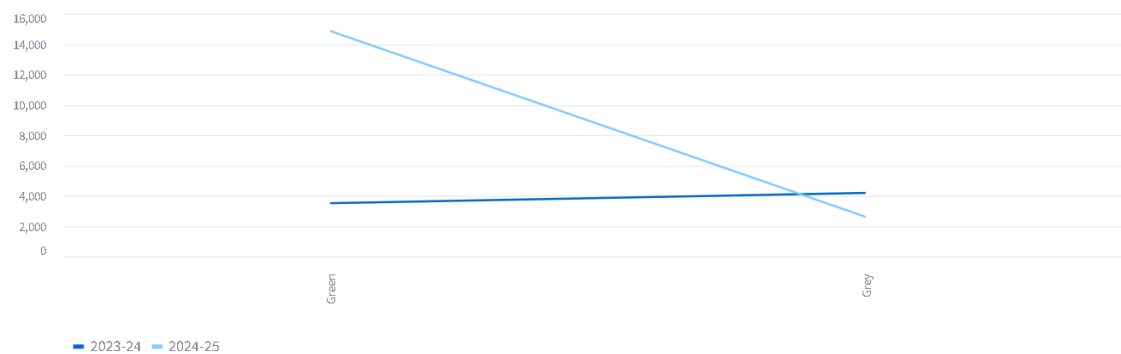
Share of houses (grey falling, green rising, as a fraction of all houses within radius)



Adoption over time

Year	Grey (non-EV)	Green (EV adopted)
2023-24	4178	3501
2024-25	2600	14866

Absolute counts (houses within each cycle's radius)



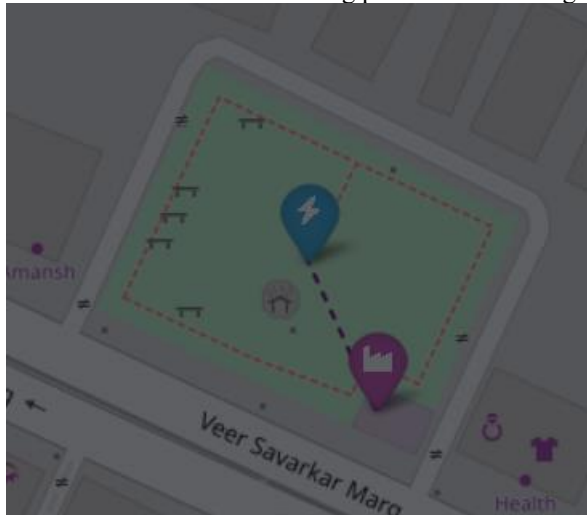
Charging station cost over time

Charging station placed at the nearest open area to the nearest substation, **0.692 km** from the locality's geocoded center.

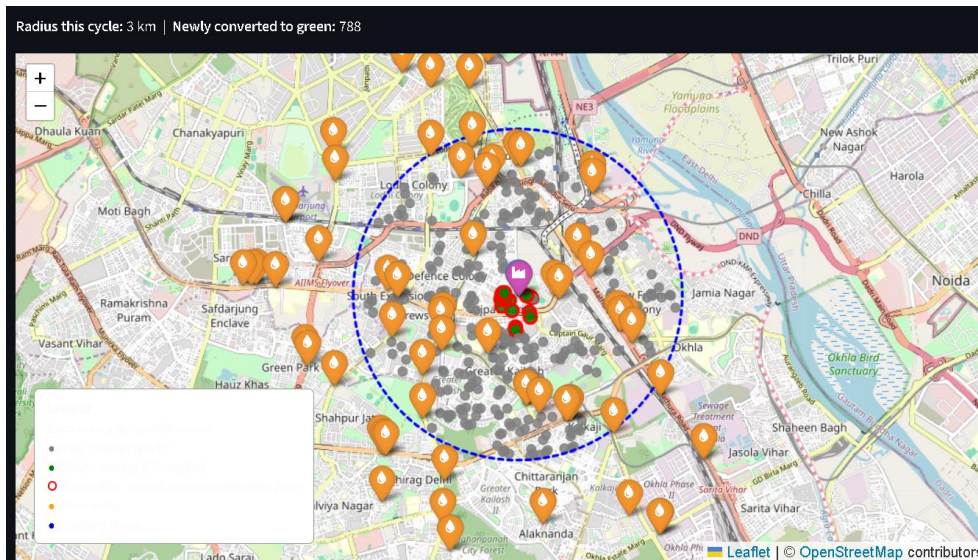
- Cabling cost: ₹181,890 (0.065 km from substation to station @ ₹2,800/m)
- Plus base setup cost: ₹247,000
- Plus station hardware cost: ₹3,300,000
- **One-time setup total: ₹3,728,890**
- Plus ₹1,300,000 expansion cost (for increasing range) added every year

5.2 Case Study: Lajpat Nagar

The density as shown of households in this region seems extremely dense as compared to other simulations, the synthetic distribution in the starting provides for a charging station in the given location.



The map shows that petrol pumps are not as dense as other locations, proving this location as one of the prime options for investment.



Adoption over time

Year	Grey (non-EV)	Green (EV adopted)
2023-24	10934	3744
2024-25	12690	15489

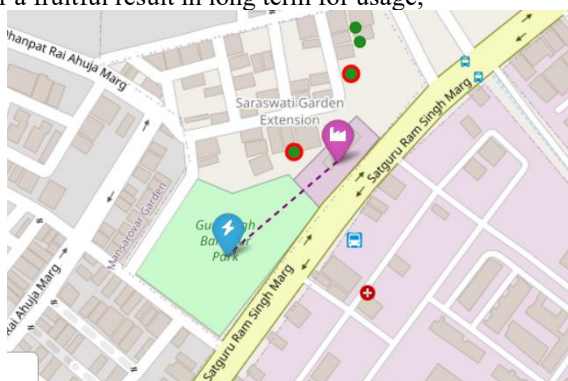
Absolute counts (houses within each cycle's radius)



5.3 Case Study: Rajouri Garden

Found 29093 houses and 38 petrol pumps in range. Charging station placed at the nearest open area to the nearest substation: 28.64421, 77.13737 (1.530 km from the geocoded center, 0.137 km of cable run from the substation).

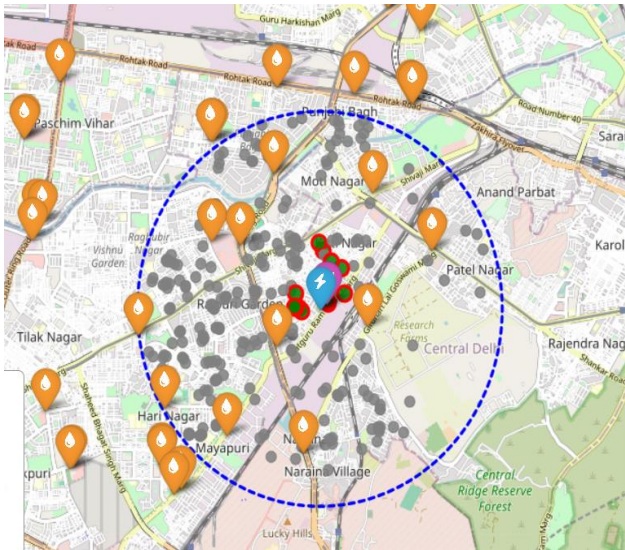
The house counts for simulation in this particular case came out to incredibly dense and high in number, allowing for a fruitful result in long term for usage,



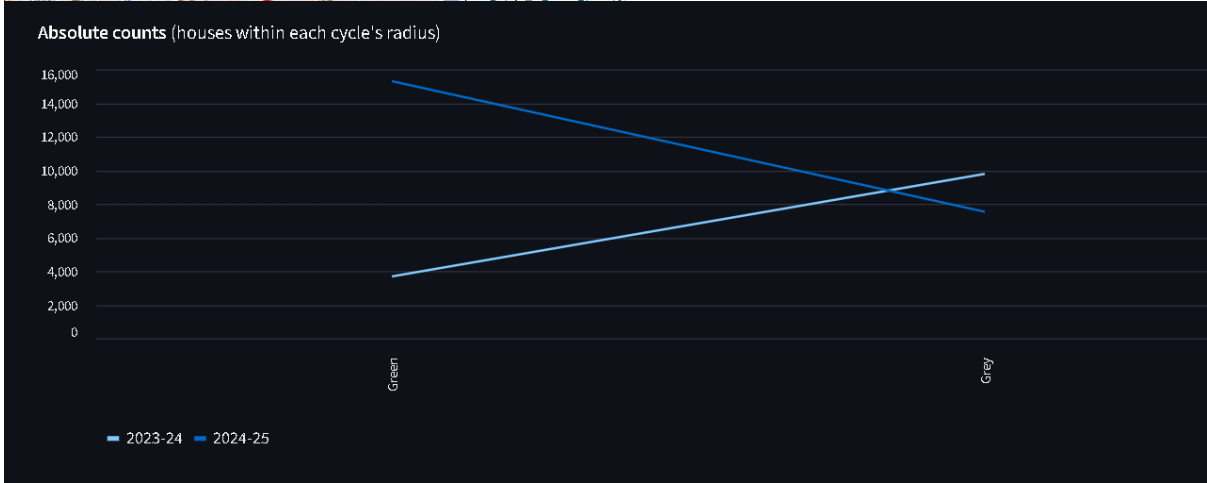
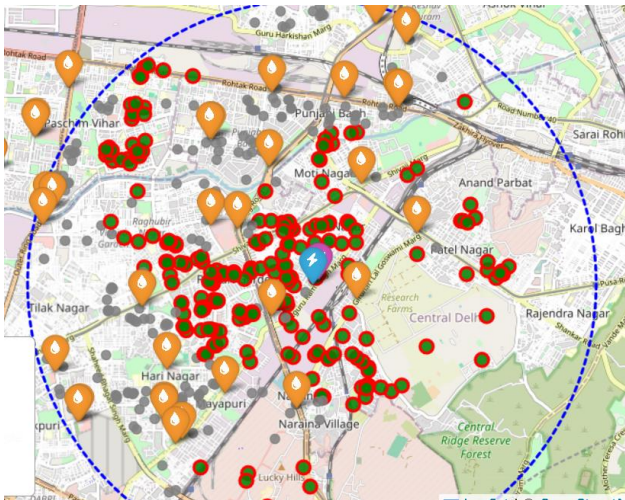
The charging station placement came out to be the most efficient in terms of the distance from the substation.

In this case there seems to be a very major change in the two years of solutions making both maps significant in terms of usage:

Year 1;



Year 2:



Cost breakdown by year			
Year	Expansion cost this year (Rs)		Cumulative cost / maintenance increment (Rs)
1		1300000	5231222.8389
2		1300000	6531222.8389

5.4 Case Study: Chandni Chowk

The area of Chandni chowk is extremely dense and hard to find space for placement in, here the majorly decisive pathway followed will be for an open area. Hence making it coincide with a region near a mall.

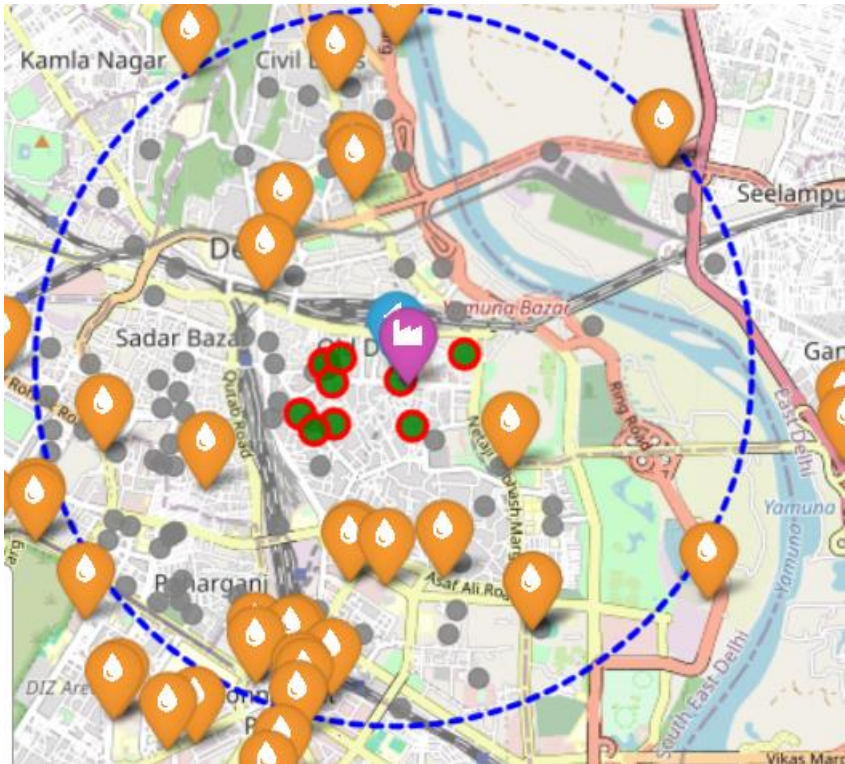


Not a lot of households recognized here, there seems to be a lot of commercial outlets near the central region of this particular area

Adoption over time

Year	Grey (non-EV)	Green (EV adopted)
2023-24	3128	1527
2024-25	7567	8319

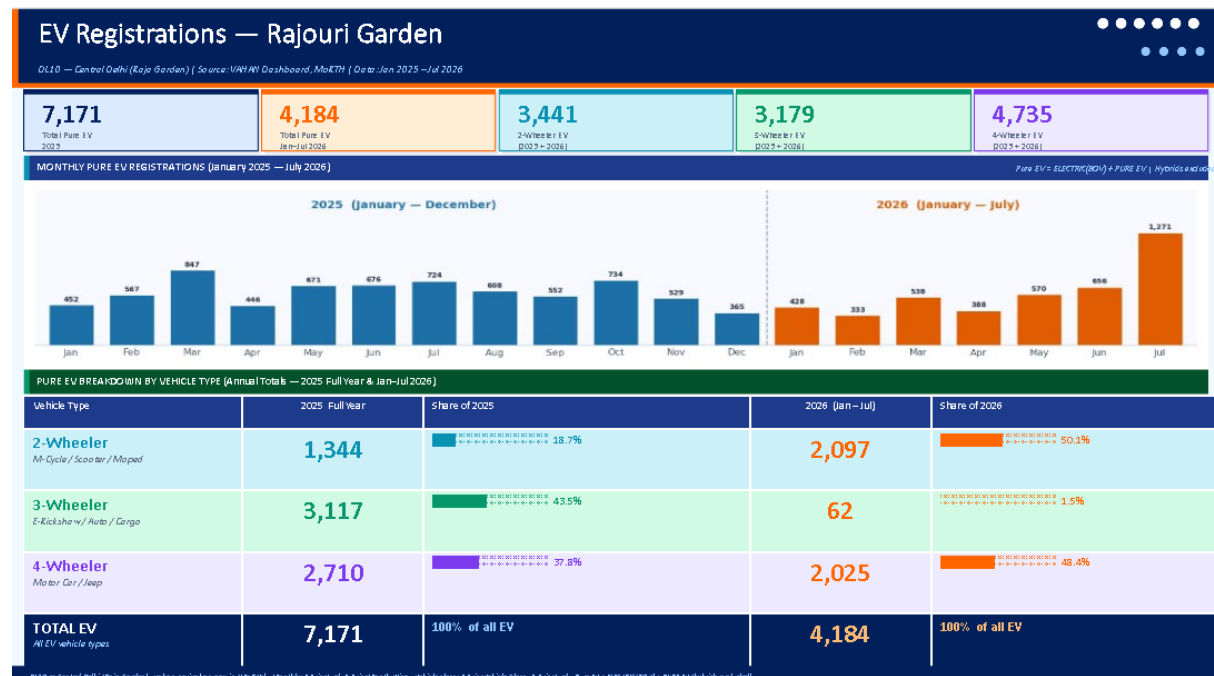
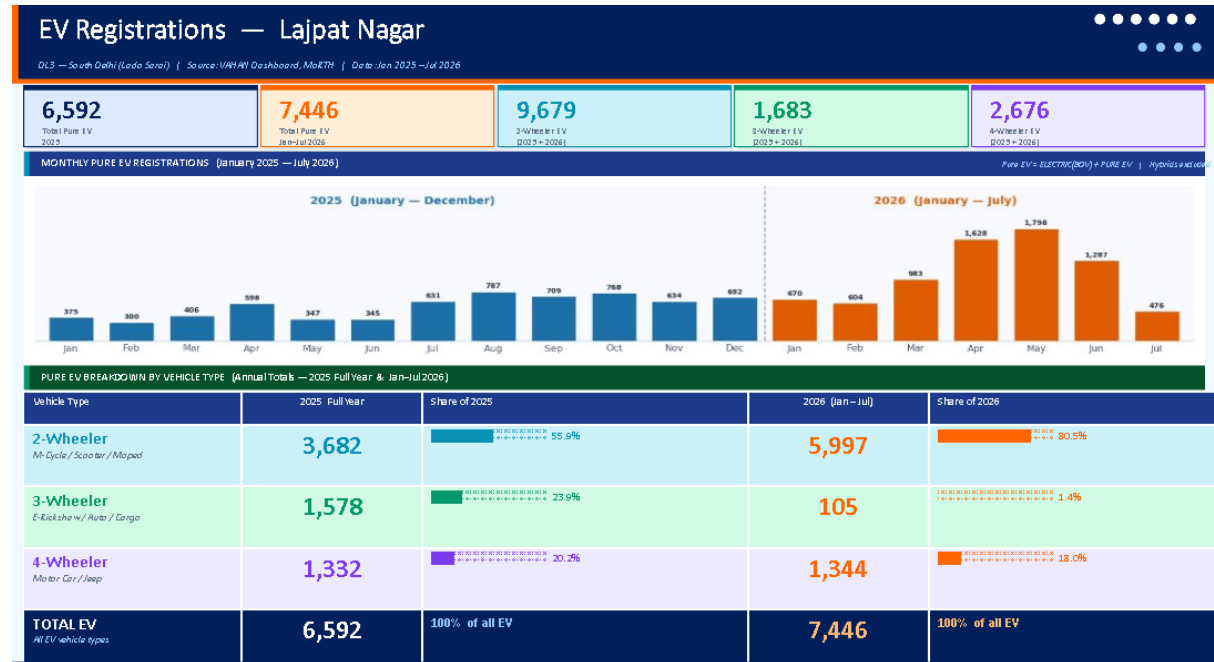
Absolute counts (houses within each cycle's radius)

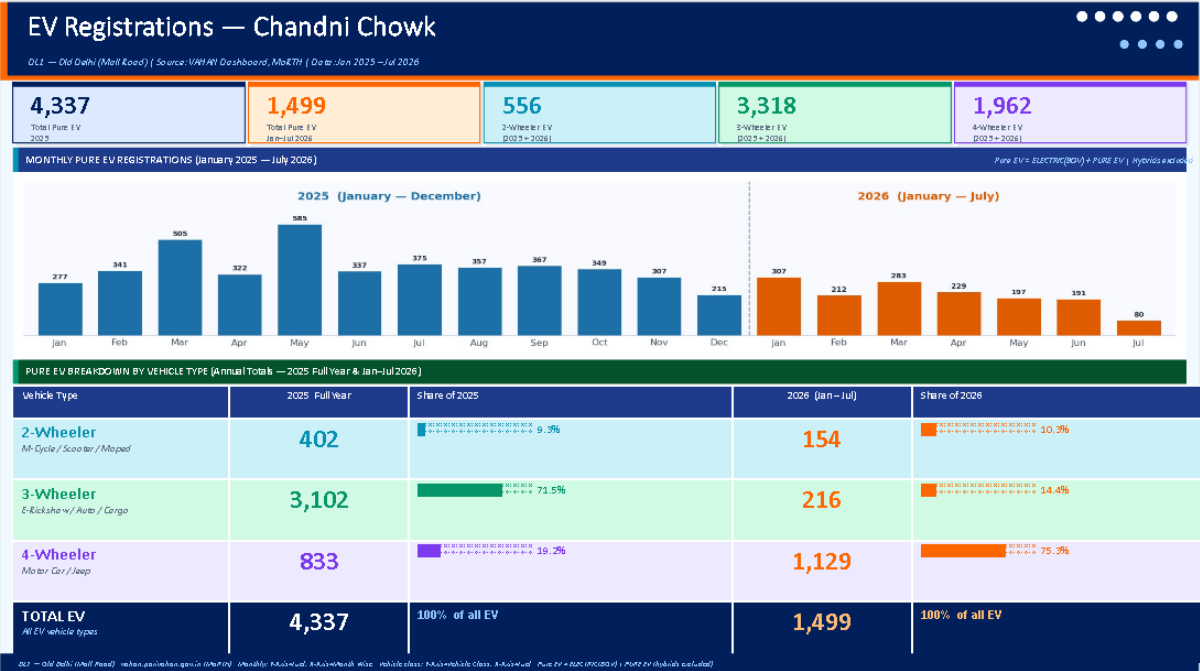


Cost breakdown by year

Year	Expansion cost this year (Rs)	Cumulative cost / maintenance increment (Rs)
1	1300000	5321123.8055
2	1300000	6621123.8055

5.5 Comparative Discussion





Across the three localities, both the simulated geospatial outcomes and the real-world VAHAN data reveal highly distinct adoption profiles driven by local geographic and demographic realities.

Rajouri Garden demonstrates the highest capital efficiency in the simulation due to minimal substation cabling requirements, which pairs effectively with its strong, sustained real-world adoption of 4-wheelers (4,735 combined units across 2025 and mid-2026). Conversely, Lajpat Nagar exhibits explosive recent growth, overtaking its entire 2025 total in just the first seven months of 2026 (7,446 total units), an acceleration driven almost entirely by the 2-wheeler segment.

Chandni Chowk remains structurally distinct; it is limited by dense commercial zoning in the simulation and dominated by 3-wheeler registrations (3,318 combined units over the recorded period) in reality. Consequently, the data proves that infrastructure deployment strategies cannot be uniform. Capital allocation must be dynamically targeted to accommodate the specific vehicle class dominance, physical space constraints, and localized cabling costs inherent to each distinct neighborhood.

6 Conclusion

The primary objective of this simulation dashboard is to serve as a predictive, capital-optimization tool for infrastructure investors, Distribution System Operators (DSOs), and Charge Point Operators (CPOs). Because large-scale EV infrastructure deployment is fundamentally constrained by available capital, the transition to electric mobility requires extreme precision in where funding is allocated.

This dashboard bridges the gap between raw geographic data and actionable financial strategy by providing a clear, data-driven methodology for investment:

- **Maximizing Conversion per Rupee:** By inputting a fixed capital budget, stakeholders can dynamically simulate and compare various urban localities side-by-side. The model weighs the location-dependent deployment costs—chiefly driven by the required cabling distance from the nearest electrical substation—against the resulting demand generation (the sheer volume of GREY households converting to GREEN).
- **Identifying High-Yield "Best Bets":** The simulation highlights the exact geographic zones that offer the highest absolute number of EV conversions for the lowest relative investment. It cuts through standard density metrics to find optimal localities where physical infrastructure friction is minimal and the financial likelihood of consumer transition is exceptionally high.
- **Defining Investment Priorities:** Once the simulation isolates these highly efficient locations, they are immediately established as the top-priority targets for real-world funding and deployment.

Ultimately, rather than relying on uniform or reactive expansion strategies, this tool ensures that every unit of capital is deployed with maximum leverage. It guarantees that initial investments are funneled exclusively into the most cost-efficient and demand-heavy areas first, accelerating the broader transition to electric vehicles while strictly safeguarding long-term operator ROI.

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